Week 9

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the linear approximation of a function f at the point $(x + \epsilon d)$ is $f(x) = \epsilon d^T \nabla f(x)$

Convex Functions

Functions of one variable

The second-order partial derivative (double derivative) is non-negative.

Functions of two variables

$$\text{Hessian Matrix} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$$

The determinant of the hessian matrix should is non-negative and $f_{xx} > 0.$

Finding minimum/ maximum points

Step 1: List the objective function $f(x)$ and $g(x)$.

Step 2: Solve $\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$ to get values of x, y and z in terms of λ
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Step 3: Substitute these values in g(x) to find critical points.